Low-Complexity Double EP Approximation for Parallel Detection-Decoding in MIMO Systems

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***Abstract*—In this letter, we propose new expectation propagation (EP)-based iterative detection and decoding (IDD) algorithms. The proposed algorithms can replace all exponential operations and eliminate the matrix inversions in both inner and outer loops, reducing significantly the computational complexity. In addition to complexity reduction, the proposed algorithm decouples the data dependency between inner loops and outer loops. Consequently, we propose the parallel EP-IDD framework to enable simultaneous iterations of detection and decoding. Simulation results demonstrate that the proposed algorithms can outperform the minimum mean-squared error (MMSE)-based IDD algorithm by 0.55 dB at the bit error rate of 10^{-3} with less computational complexity consumption. Compared with the state-of-the-art EP-based IDD algorithms, the proposed algorithm achieves a 69.55% to 95.27% reduction in computational complexity without noticeable performance loss.**

***Index Terms*—Expectation propagation (EP), parallel iterative detection and decoding (IDD), multiple-input multiple-output (MIMO), polar codes.**

I. Introduction

To CATER to a diverse range of usage scenarios, the fifth-generation (5G) wireless communication system should evolve cutting-edge technologies, bringing a revolutionary breakthrough in terms of data rates, spectral efficiency (SE), latency, and reliability. Among the leading technologies for 5G systems, massive multiple-input multiple-output (MIMO) detection and channel coding are two key mainstays in the physical layer. Massive MIMO aims to provide high throughput, reliability and SE. Advanced channel codes serve as critical roles in achieving capacity in wireless systems.

In MIMO communication systems, minimum mean-squared error (MMSE) detection has been widely applied as a classic linear detection. However, the linear detection suffers from performance-complexity bottlenecks as the number of antennas and modulation order grows rapidly. The limitation has spurred research into nonlinear detection algorithms [1], [2], [3]. Among non-linear detectors, expectation propagation (EP) algorithm [1] exhibits near-optimal performance across various antenna and modulation configurations. To reduce the complexity, modified EP schemes have been proposed to approximate matrix inversion [4], [5], [6] and support the practical mobile communication applications [7]. To achieve better performance, EP detection can be integrated with the iterative belief propagation (BP) decoding algorithms for iterative detection and decoding (IDD).

Compared with MMSE-based IDD strategies [8], [9], [10], EP-based IDD achieves superior performance [11], [12], [13] and strong robustness in multi-user systems [14]. The block EP (BEP) is firstly applied to EP-IDD systems in [11]. Rather than direct moment calculation in BEP, double EP (DEP) scheme in [12] considers moment matching (MM) in both loops for a better posterior estimation. To reduce the excessive complexity, the EP approximation IDD scheme (EPA-IDD) [13] adopts approximate computations in the inner loop. Unfortunately, all aforementioned EP-IDD strategies with likelihoods involve considerable exponential multiplication operations for implementation. Besides, the repeated large dimension matrix inversions also lead to an increase in computational complexity.

In this letter, we propose a EP-IDD approximation scheme, named double EPA (DEPA), to eliminate the matrix inversion in both loops. Moreover, the proposed scheme formulates the inner and outer loops computations in log-likelihood (LL) and LL ratio (LLR) domain. The excessive computation caused by matrix inversions for each loop and exponential operations in likelihoods can be removed. Besides, we propose a parallel EP-IDD framework to decouple the data dependency between inner loops and outer loops for throughput improvement.

Contribution: In this letter, we introduce a low-complexity DEPA scheme for parallel detection and decoding. The proposed algorithm replaces all exponential operations in likelihoods to achieve the MM procedure of IDD. Furthermore, the scheme eliminates the matrix inversions in both loops through posterior update approximation. This results in a evident reduction in computational complexity compared with the state-of-the-art (SOA) DEP detector. Finally, we propose a parallel EP-IDD framework for our IDD scheme to achieve iteration convergence with minimal performance loss.

II. Preliminaries

A. MIMO System Model

We consider a coded uplink MIMO system with N_t transmitting antennas and N_r receiving antennas, $N_r > N_t$. On the transmitter side, the information vector $\boldsymbol{a} = [a_1, \ldots, a_K]$ is encoded into the codeword $\boldsymbol{b} = [b_1, \ldots, b_N]$ as a code

Received 26 October 2024; revised 8 December 2024; accepted 12 December 2024. Date of publication 6 May 2025; date of current version 10 November 2025. This work was supported in part by NSFC under Grant 62122020, Grant 62331009, and Grant 62471135; in part by the Jiangsu Provincial NSF under Grant BG2024004 and Grant BM2023016; and in part by the Fundamental Research Funds for the Central Universities. The associate editor coordinating the review of this article and approving it for publication was H. Jung. *(Corresponding author: Chuan Zhang.)*

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Digital Object Identifier 10.1109/LWC.2024.3520556

rate $R = K/N$. After interleaving, the codeword is divided into blocks of Q bits and modulated into the transmitting symbol vector $\boldsymbol{x}$, whose values are taken from a complex-valued M-ary constellation set $\tilde{\mathcal{A}}$, $M = |\tilde{\mathcal{A}}| = 2^Q$. The symbol vector is partitioned into P sets of length N_t, $\tilde{\boldsymbol{x}} = [\tilde{\boldsymbol{x}}(1), \ldots, \tilde{\boldsymbol{x}}(P)]$. Each set is distributed into N_t substreams and transmitted over the i.i.d. Rayleigh channel $\tilde{\mathbf{H}} \in \mathbb{C}^{N_r \times N_t}$. Hereafter, the set index is omitted for simplified notation as $\tilde{\boldsymbol{x}} = [\tilde{x}_1, \ldots, \tilde{x}_{N_t}]^T \in \mathbb{C}^{N_t \times 1}$, where $[\ ]^T$ denotes the transpose operation. On the receiver side, the received symbol vector $\tilde{\boldsymbol{y}} = [\tilde{y}_1, \ldots, \tilde{y}_{N_r}]^T \in \mathbb{C}^{N_r \times 1}$ can be expressed as $\tilde{\boldsymbol{y}} = \tilde{\mathbf{H}}\tilde{\boldsymbol{x}} + \tilde{\boldsymbol{n}}$, where $\tilde{\boldsymbol{n}}$ is a complex-valued additive white Gaussian noise (AWGN) vector that $\tilde{\boldsymbol{n}} \sim \mathcal{CN}(\mathbf{0},\ \sigma_{\tilde{n}}^2\mathbf{I})$.

In practice, the above complex-valued system model is transformed into the real-valued domain by real value decomposition [1].

$$\boldsymbol{y} = \mathbf{H}\boldsymbol{x} + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{y} \in \mathbb{R}^{2N_r \times 1}$, $\mathbf{H} \in \mathbb{R}^{2N_r \times 2N_t}$, $\boldsymbol{x} \in \mathbb{R}^{2N_t \times 1}$, and $\boldsymbol{n} \sim \mathcal{N}(\mathbf{0},\ \sigma_n^2\mathbf{I}) \in \mathbb{R}^{2N_r \times 1}$. After the reformulation, the symbols of $\boldsymbol{x}$ are real-valued points in the constellation set $\mathcal{A}$ with $|\mathcal{A}| = \sqrt{M}$, where $\mathcal{A} = [-\sqrt{M}+1, \ldots, -1, +1, \ldots, \sqrt{M}-1]$ with the average symbol energy E_s.

The posterior probability of the transmitting symbol vector $\boldsymbol{x}$ given the receiving symbol vector $\boldsymbol{y}$ yields

$$p(\boldsymbol{x}|\boldsymbol{y}) \ = \frac{p(\boldsymbol{y}|\boldsymbol{x})p_D(\boldsymbol{x})}{p(\boldsymbol{y})} \propto \mathcal{N}\Big(\boldsymbol{y}{:}\mathbf{H}\boldsymbol{x}, \sigma_n^2\mathbf{I}\Big) \prod_{i=1}^{2N_t} p_D(x_i), \tag{2}$$

where the prior probability $p_D(\boldsymbol{x})$ is a discrete distribution.

B. EP-Based Detection With IDD Schemes

According to the aforementioned posterior model (2), EP algorithm provides feasible Gaussian prior approximation terms $t(x_i)$ to iteratively estimate the product of the non-Gaussian terms $p_D(x_i)$ in (2). Therefore, the Gaussian approximation posterior $q^{[l]}(\boldsymbol{x})$ can be factorized as

$$q^{[l]}(\boldsymbol{x}) \sim \mathcal{N}\Big(\boldsymbol{x} : \boldsymbol{\mu}^{[l]}, \boldsymbol{\Sigma}^{[l]}\Big) \propto \mathcal{N}\big(\boldsymbol{y} : \mathbf{H}\boldsymbol{x}, \sigma_n^2\mathbf{I}\big) \prod_{i=1}^{2N_t} t^{[l]}(x_i), \tag{3}$$

where $t^{[l]}(x_i)$ is an unnormalized Gaussian term in the l-th iteration, denoted as $t^{[l]}(x_i) = e^{\gamma_i^{[l]} x_i - \frac{1}{2}\Lambda_i^{[l]} x_i^2}$. The mean $\boldsymbol{\mu}^{[l]}$ and covariance $\boldsymbol{\Sigma}^{[l]}$ of $q^{[l]}(\boldsymbol{x})$ can be computed

$$\boldsymbol{\Sigma}^{[l]} = \Big(\mathbf{A} + \boldsymbol{\Lambda}^{[l]}\Big)^{-1},\ \boldsymbol{\mu}^{[l]} = \boldsymbol{\Sigma}^{[l]}\Big(\mathbf{b} + \boldsymbol{\gamma}^{[l]}\Big), \tag{4}$$

where $\mathbf{A} = \sigma_n^{-2}\mathbf{H}^T\mathbf{H}$, $\mathbf{b} = \sigma_n^{-2}\mathbf{H}^T\boldsymbol{y}$, and the prior parameters $\boldsymbol{\gamma}^{[l]} = [\gamma_1^{[l]}, \ldots, \gamma_{2N_t}^{[l]}]^T$, $\boldsymbol{\Lambda}^{[l]} = \mathrm{diag}([\Lambda_1^{[l]}, \ldots, \Lambda_{2N_t}^{[l]}])$. Therefore, the probability density function (PDF) $q^{[l]}(\boldsymbol{x})$ can be confirmed as (4). In order to approximate (2) with (3) in each iteration, we denote a i-th marginal $q^{[l]}(x_i)$ and its cavity marginal $q^{\backslash i[l]}(x_i)$, where $q^{[l]}(x_i) \sim \mathcal{N}(x_i : \mu_i^{[l]}, \Sigma_{ii}^{[l]})$, $\Sigma_{ii}^{[l]}$ is the i-th diagonal element of $\boldsymbol{\Sigma}^{[l]}$, and $q^{\backslash i[l]}(x_i)$ is denoted as

$$q^{\backslash i[l]}(x_i) = q^{[l]}(x_i)/t^{[l]}(x_i) \sim \mathcal{N}\Big(x_i : \mu_{E_i}^{[l]}, \sigma_{E_i}^{2[l]}\Big). \tag{5}$$

According to (3), (4), the mean $\mu_{E_i}^{[l]}$ and the variance $\sigma_{E_i}^{2[l]}$ of $q^{\backslash i[l]}(x_i)$ can be calculated as

$$\sigma_{E_i}^{2[l]} = \frac{\Sigma_{ii}^{[l]}}{\Big(1 - \Sigma_{ii}^{[l]}\Lambda_i^{[l]}\Big)},\quad \mu_{E_i}^{[l]} = \sigma_{E_i}^{[l]}\left(\frac{\mu_i^{[l]}}{\Sigma_{ii}^{[l]}} - \gamma_i^{[l]}\right). \tag{6}$$

For the next iteration, we consider MM between the Gaussian approximation PDF with $t^{[l+1]}(x_i)$ in (3) and true PDF in (2) based on Kullback-Leibler divergence minimization principle.

$$\hat{p}(x_i) = \underbrace{q^{\backslash i[l]}(x_i)p_D(x_i)}_{\left(\mu_{p_i}^{[l]}, \sigma_{p_i}^{2[l]}\right)} \overset{\text{MM}}{\Longleftrightarrow} \underbrace{q^{\backslash i[l]}(x_i)}_{\left(\mu_{E_i}^{[l]}, \sigma_{E_i}^{2[l]}\right)}\ e^{\gamma_i^{[l+1]} x_i - \frac{1}{2}\Lambda_i^{[l+1]} x_i^2}. \tag{7}$$

Based on (7), the parameter pair $(\gamma_i^{[l+1]}; \Lambda_i^{[l+1]})$ can be obtained by the first two moments pair $(\mu_{p_i}^{[l]}, \sigma_{p_i}^{[l]}; \mu_{E_i}^{[l]}, \sigma_{E_i}^{[l]})$ to update the mean $\boldsymbol{\mu}^{[l+1]}$ and the covariance $\boldsymbol{\Sigma}^{[l+1]}$ of $q^{[l+1]}(\boldsymbol{x})$ as (4) for next iteration.

To simplify the posterior updating, a marginal cavity approximation in [4] can be redefined as $q^{\backslash i}(x_i) = e^{\rho_i x_i - \frac{1}{2}V_i x_i^2}$. The update equations for $(\rho_i; V_i)$ can be expressed as

$$\boldsymbol{\rho} = \mathbf{H}^T\boldsymbol{m}/\sigma_n^2 + \boldsymbol{V}\boldsymbol{\mu}_p,\ \boldsymbol{m} = \boldsymbol{y} - \mathbf{H}\boldsymbol{\mu}_p. \tag{8}$$

Considering the channel hardening [4], the parameter $\boldsymbol{V}$ in (8) can be approximated as $\mathrm{diag}(\mathbf{H}^T\mathbf{H})/\sigma_n^2$.

To efficiently ascertain the parameters $(\mu_{p_i}^{[l]}, \sigma_{p_i}^{[l]})$, it is necessary to consider decoding to update the prior $p_D(\boldsymbol{x})$ in detection, and the detailed steps vary between separated detection-decoding (SDD) and IDD.

Generally, the prior $p_D(\boldsymbol{x})$ is set to a discrete uniform distribution under the SDD scheme, i.e., $p_D(x_i) = \frac{1}{|\mathcal{A}|}\mathbb{I}_{\mathcal{A}}(x_i)$, $\mathbb{I}_{\mathcal{A}}(x)$ is the indicator function that yields '1' when $x \in \mathcal{A}$, and '0' otherwise. The decoder in the IDD receiver can provide extrinsic message $p_d(\boldsymbol{x})$ to refine the parameter pair $(\gamma_i; \Lambda_i)$ of $p_D(\boldsymbol{x})$ for detection. In [11], the BEP scheme projects $p_d^{[l+1]}(x_i)$ into the prior Gaussian term $t^{[l+1]}(x_i)$ as

$$\frac{\gamma_i^{[l+1]}}{\Lambda_i^{[l+1]}} = \mathbb{E}_{p_d^{[l+1]}}(x_i),\quad \frac{1}{\Lambda_i^{[l+1]}} = \mathbb{E}_{p_d^{[l+1]}}\left[\Big(x_i - \mathbb{E}_{p_d^{[l+1]}}(x_i)\Big)^2\right]. \tag{9}$$

In [12], the DEP scheme applies the MM method and the $p_d(\boldsymbol{x})$ from decoder for posterior updating instead of the uniform distribution $p_D(\boldsymbol{x})$ in (7). Based on (7), the parameter pair $(\gamma_i; \Lambda_i)$ can be updated as

$$\Lambda_i^{[l+1]} = \frac{1}{\sigma_{p_i}^{2[l]}} - \frac{1}{\sigma_{E_i}^{2[l]}},\quad \gamma_i^{[l+1]} = \frac{\mu_{p_i}^{[l]}}{\sigma_{p_i}^{2[l]}} - \frac{\mu_{E_i}^{[l]}}{\sigma_{E_i}^{2[l]}}. \tag{10}$$

The existing EP-IDD schemes [13] to propose adaptive outer-to-inner MM methods. However, limited by inevitable matrix inversion and nonlinear operations in each loop, the overall complexity of IDD scheme will be improved significantly as the number of iterations increases.

III. Low-Complexity Double EPA Detector

The equations of (4)-(10) summarize the core concept of EP algorithm and its IDD schemes. The key challenges of the existing EP-IDD schemes into the following two main aspects: 1) The repeated large dimension matrix inversions in the inner loop of detection iteration and outer loops as (4) lead to an increase in computational complexity. 2) The detection and decoding iterations in the inner loops need to alternate in sequentially updating the moments $(\boldsymbol{\mu}; \boldsymbol{\Sigma})$, resulting in a reduction in update efficiency.

To simplify the operations and facilitate efficient message exchange within the IDD loops, we propose the DEPA algorithm to reorganize the data flow of the DEP algorithm

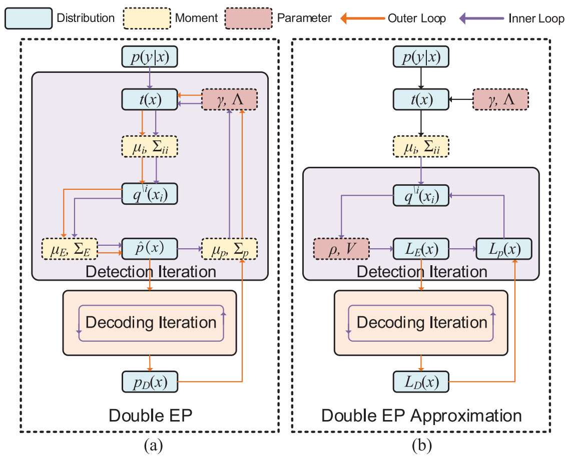


Fig. 1. (a) IDD receiver diagram of DEP algorithms. (b) IDD receiver diagram of DEPA algorithms.

as Fig. 1. By reorganizing the data flow within the inner and outer loops, the proposed algorithm significantly reduces computational complexity and eliminates the need for matrix inversions in both loops. Besides, we can further discuss how to decouple the data dependency between inner and outer loops to propose a parallel EP-IDD framework.

A. Double EP Approximation

Compared with the prior parameter pair $(\boldsymbol{\gamma};\boldsymbol{\Lambda})$ updating in both IDD loops for $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ convergence [11], [12], we assume the μ_{p_i} and $\sigma^2_{p_i}$ in (7) can be consistent with μ_i and Σ_{ii} at the convergence point, i.e., $\mu_i = \mu_{p_i}$, $\Sigma_{ii} = \sigma^2_{p_i}$. Therefore, the DEPA algorithm can focus on updating the $(\boldsymbol{\mu}_p, \boldsymbol{\sigma}^2_p)$ rather than the parameter pair $(\boldsymbol{\gamma};\boldsymbol{\Lambda})$.

By omitting the update of the parameter pair $(\boldsymbol{\gamma},\boldsymbol{\Lambda})$, the proposed DEPA algorithm allocates updates for these two terms in (7) to the inner and outer loops, respectively. The corresponding updated messages iteratively interact with each loop as extrinsic messages. To perform the computations in the LL domain, the equation of (7) can be rewritten as

$$\mathsf{L}_p(x) = \mathsf{L}_E(x) + \mathsf{L}_D(x), \tag{11}$$

where L_p, L_E, and L_D denote the symbol-based LL messages for posterior, extrinsic, and prior probabilities, respectively.

In the detection iteration, to obtain the extrinsic LL messages L_E from EPA detection, we derive the extra mean $\hat{\boldsymbol{\mu}}_E$ from (8) as follows.

$$\hat{\mu}^{[l]}_{E_i} = \rho_i / V_i = \left(\mathrm{b}_i - \sum_{j=1}^{2N_t} \mathrm{A}_{ij}\, \mu^{[l-1]}_{p_i} \right) / V_i + \mu^{[l-1]}_{p_i}. \tag{12}$$

The extra variance $\hat{\boldsymbol{\sigma}}^2_{E_i}$ fixed equals to $1/V_i$. We can achieve μ_p computations using all $\sqrt{M}$ constellation points in $\mathcal{A}$ with likelihoods. Besides, we also propose a minimal version DEPA (mDEPA) scheme for μ_p computations. To avoid the complexity grows rapidly when M increases, we employ the maximum selection L_p within the constellation set. Therefore, the μ_p in (12) can be simplified as

$$\mu^{[l]}_p = \arg\max_{s_k \in \mathcal{A}} \mathsf{L}^{[l]}_p(x = s_k). \tag{13}$$

With the fixed extra variance σ^2_E, the extrinsic LL messages L_E from detection are defined as

$$\mathsf{L}_E(x) = -\frac{|x - \hat{\mu}_E|^2}{2\hat{\sigma}^2_E},\ x \in \mathcal{A}. \tag{14}$$

In the decoding iteration, the symbol-based LL messages $\mathsf{L}_E(x)$ from detection can be approximately converted to bit-based LLR message $\mathsf{L}_{\tilde{E}}(x^j)$ and deinterleaved before being provided to decoders, as shown in (15).

$$\mathsf{L}_{\tilde{E}}\left(x^j\right) \simeq \frac{\sum_{s_k \in \mathcal{A}^0_j} \mathsf{L}_E(x = s_k)}{\sum_{s_k \in \mathcal{A}^1_j} \mathsf{L}_E(x = s_k)}, \tag{15}$$

where x^j denotes the *j*-th bit of the symbol *x*, and $\mathcal{A}^b_j$ denotes the subset of $\mathcal{A}$ where the *j*-th bit of these symbols is *b*.

After decoding iterations, the decoder outputs the updated LLR messages $\mathsf{L}_{\tilde{D}}(x^j)$ and computes their extrinsic messages feedback to the detector. The bit-based extrinsic message from decoder can be computed as $\Delta\mathsf{L}_{\tilde{D}}(x^j) = \mathsf{L}_{\tilde{D}}(x^j) - \mathsf{L}_{\tilde{E}}(x^j)$. Inspired by the theorem in [15], we can accumulate the bit belief to achieve the bit-to-symbol belief conversion. Hence, the symbol-based prior LL messages from decoder $\mathsf{L}_D(x)$ can be defined as

$$\mathsf{L}_D(x) = \sum_{j=1}^{\log_2 \sqrt{M}} \ln\left(1 + e^{(1-2x^j)\cdot \Delta\mathsf{L}_{\tilde{D}}(x^j)}\right). \tag{16}$$

To approximate the terms in (16), we provide the simplified form of $\mathsf{L}_D(x)$ as

$$\mathsf{L}_D(x) \simeq \sum_{j=1}^{\log_2 \sqrt{M}} \left|\Delta\mathsf{L}_{\tilde{D}}\left(x^j\right)\right| \cdot \mathbb{I}_{\mathcal{B}}\left(x^j\right), \tag{17}$$

where $\mathcal{B} = \{x^j | x^j \neq \frac{1}{2}[1 + \mathrm{sign}(\Delta\mathsf{L}_{\tilde{D}}(x^j))]\}$.

By executing the $\mathsf{L}_E(x)$ and $\mathsf{L}_D(x)$ updates in the detection and decoding iterations, respectively, the DEPA algorithm shortens the operational dataflow excluding matrix inversions in both loops. Moreover, all exponential computations in the data flow are eliminated in the LL and LLR domain. Additionally, unlike the data dependency within IDD in the DEP algorithm, the DEPA algorithm decouples the processing of both loops. Consequently, we can develop a parallel IDD framework based on the DEPA algorithm.

B. Parallel Double EP Approximation

Regarding the existing EP-IDD solutions, the IDD schemes adopt the turbo principle model [11], [12], [13] for belief updating shown in Fig. 1(a). Limited by the dependency between the inner and outer loops, the nature of serial updates results in a reduction in convergence speed, posing challenges for achieving high throughput and low latency in MIMO systems.

As the iterations of L_E and L_D are performed independently and do not require alternating updates of the moments $(\boldsymbol{\mu};\boldsymbol{\Sigma})$, we apply the DEPA algorithm to the parallel IDD framework for MIMO systems. Fig. 1(b) shows the parallel IDD model for the DEPA algorithm. Initialized identically by $\mu^{[0]}_{p_i} = \mu^{[0]}_i$ and $\sigma^{2[0]}_{p_i} = \Sigma^{[0]}_{ii}$ as (4), the detection and decoding iterations can simultaneously perform symbol-based and bit-based belief message updates separately. Subsequently, the

Algorithm 1: The Procedure of DEPA Algorithms

Input: $\boldsymbol{y}$, $\mathbf{H}$, $\boldsymbol{\gamma}$, $\mathbf{\Lambda}$, σ_n^2.
Initialization: $\mathbf{A} = \sigma_n^2(\mathbf{H}^T\mathbf{H} + \mathbf{\Lambda})^{-1}$, $\mathbf{b} = \sigma_n^{-2}\mathbf{H}^T\boldsymbol{y}$. $L_D^{[0]}(\boldsymbol{x}) = 0$.
Compute $\mu_{p_i}^{[0]} = \mu_i^{[0]}$ and $\sigma_{p_i}^{2[0]} = \Sigma_{ii}^{[0]}$ with (4) for all i.
for $l = 0, \dots, L-1$ **do**
  Parallel Interacting Variables:
  $\mathsf{L}_E(x)$, $\mathsf{L}_D(x)$.
  Parallel Detection Loop:
  for $i = 1, \dots, 2N_t$ **do**
    Compute the extra mean $\hat{\mu}_{E_i}^{[l+1]}$ with $\mu_{p_i}^{[l]}$ as (12).
    Update the extrinsic LLRs from detector $\mathsf{L}_E^{[l+1]}(x_i)$, $x_i \in \mathcal{A}$ as (14).
    Update the posterior LLRs $\mathsf{L}_p^{[l+1]}(x)$ as (11).
    Compute the mean $\mu_{p_i}^{[l+1]}$ computation (DEPA), or maximum selection as (13) (mDEPA).
  Parallel Decoding Loop:
  if $l == 0$ **then**
    $\mathsf{L}_E^{[0]}(x) = -V(x - \hat{\mu}_p^{[0]})^2/2$.
  Compute bit-based extrinsic LLR message from detector $\mathsf{L}_{\tilde{E}}^{[l]}(x^j)$ as (15).
  After obtaining $\mathsf{L}_{\tilde{D}}^{[l]}(x^j)$ through iterative decoding, the extrinsic LLR messages from decoder: $\Delta\mathsf{L}_{\tilde{D}}^{[l+1]}(x^j) = \mathsf{L}_{\tilde{D}}^{[l]}(x^j) - \mathsf{L}_{\tilde{E}}^{[l]}(x^j)$.
  Update the symbol-based extrinsic LLR messages from decoder $\mathsf{L}_D^{[l+1]}(x)$ as (17).
Output: Decoded bit $\hat{\boldsymbol{x}}$

updated $\mathsf{L}_E(x)$ and $\mathsf{L}_D(x)$ will be provided together for posterior L_p computations in (11). Therefore, the detection and detection collaboratively update $\mathsf{L}_p(x)$ message to decouple the data dependency of inner and outer loops. The whole procedure is summarized in Algorithm 1.

To highlight the advantages of the parallel IDD framework in time complexity, we define the number of inner loops for detection, decoding iterations and outer loops as I_{det}, I_{dec} and L, respectively, and the unit time for each detection and decoding iterations can be denoted as T_{det} and T_{dec}, respectively. Compared to the total time complexity of DEP algorithm $T_{\text{DEP}} = L\cdot(I_{\text{det}}T_{\text{det}}+I_{\text{dec}}T_{\text{dec}})$, the parallel IDD framework reduce the time complexity to $L \cdot \max(I_{\text{det}}T_{\text{det}},\ I_{\text{dec}}T_{\text{dec}})$.

IV. Simulation Results and Discussions

In this section, we provide the error-rate performance of the proposed DEPA algorithms and the comparisons with other IDD schemes, such as MMSE parallel interference cancellation (MMSE-PIC), DEP, and EPA-IDD algorithms [8], [12], [13]. Additionally, the complexity analysis among these algorithms is also presented. For simulations, the coded MIMO system in this letter encodes codewords of length $N = 1024$ with $R = 1/2$ under polar codes. To preserve the low-latency feature of IDD schemes, we adopt BP polar decoding to assist the aforementioned detectors under Gaussian approximation construction [16].

A. Error Rate Performance

To comprehensively demonstrate the strengths and weaknesses of the proposed algorithms in various cases, we provide the bit error rate (BER) performance of IDD schemes under different antenna ratios and modulations.

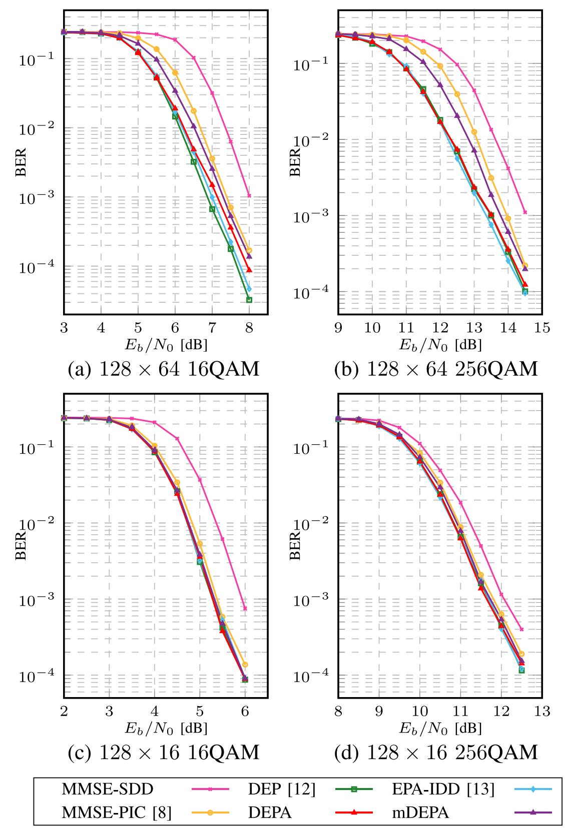


Fig. 2. Comparison results of BER performance between the proposed algorithms and existing IDD schemes.

Fig. 2 shows the BER performance of our algorithms and other IDD schemes in polar coded MIMO systems. To better illustrate their generalization, we provide two distinct cases, each exemplifying diversity in terms of antenna ratios (a) $N_r = 128, N_t = 64$, (b) $N_r = 128, N_t = 16$, modulations (a) 16QAM, (b) 256QAM, respectively. For IDD schemes, the number of inner loops for detection iterations I_{det} and decoding iterations I_{dec} is set to 2, and the number of outer loops L equals to 10. Besides, we set $I_{\text{dec}} = 20$ for MMSE-SDD algorithm to ensure a fair comparison with IDD schemes.

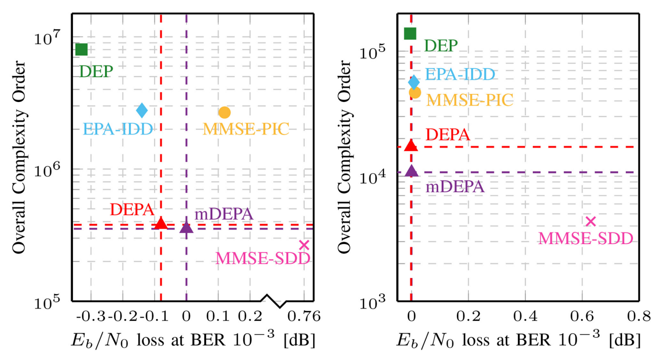
In Fig. 2, it can be observed that EP-based IDD algorithms exhibit comprehensive performance advantages over MMSE-based IDD algorithm. The proposed DEPA algorithm can achieve similar performance as other EP-based IDD schemes. Especially in the 128 × 64, 256QAM case, the DEPA outperforms the MMSE-PIC by 0.55 dB at BER $= 10^{-3}$. Besides, the mDEPA algorithm can reduce the complexity of DEPA with negligible performance loss in the 128 × 16 cases. The complexity analysis is discussed in the following section.

B. Complexity Analysis

As the total number of decoding iterations has been unified for IDD algorithms, we omit the decoding complexity and only focus on the detectors in Table I. It can be noted that the complexity of DEPA algorithm is reduced by its matrix inversion elimination in L outer loops. Moreover, the mDEPA

TABLE I
COMPLEXITY ORDER COMPARISONS FOR ALGORITHMS

Algorithm		**Complexity Order**	
		Multiplications	Exponentiations
MMSE-SDD		$N_t^3 + N_t^2$	0
MMSE-PIC [8]		$L(N_t^3 + N_t^2)$	$LN_t\lvert\mathcal{A}\rvert$
DEP [12]		$L(N_t^3 + N_t^2 + I_{\text{det}}(N_t^3 + N_t^2 + N_t\lvert\mathcal{A}\rvert))$	$LI_{\text{det}}N_t\lvert\mathcal{A}\rvert$
EPA-IDD [13]		$L(N_t^3 + N_t^2 + I_{\text{det}}(N_t^2 + N_t\lvert\mathcal{A}\rvert))$	$LI_{\text{det}}N_t\lvert\mathcal{A}\rvert$
This work	DEPA	$N_t^3 + N_t^2 + LI_{\text{det}}(N_t^2 + N_t\lvert\mathcal{A}\rvert)$	$LI_{\text{det}}N_t\lvert\mathcal{A}\rvert$
	mDEPA	$N_t^3 + N_t^2 + LI_{\text{det}}(N_t^2 + N_t\lvert\mathcal{A}\rvert)$	0



(a) $N_t = 64$ (b) $N_t = 16$

Fig. 3. Overall complexity order comparisons for the algorithms with $N_r = 128$, 16 QAM, $R = 1/2$.

algorithm can omit the exponential operations for μ_p estimation as (13). Comprehensively considering the complexity of the two algorithmic operations in Table I as [17], we denote the complexity of multiplication and exponentiation as $\mathit{Mult}(Q_{\text{word}})$ and $\log_2 Q_{\text{word}} \cdot \mathit{Mult}(Q_{\text{word}})$, respectively, where Q_{word} represents the computer word size, typically 32 bits.

Fig. 3 illustrates the trade-off between overall complexity order and BER performance for $N_r = 128$, 16QAM, and $R = 1/2$. We present examples to discuss the BER performance and complexity gains of our algorithm under different antenna ratio conditions.

When $N_t = 64$, the DEPA algorithm achieves a complexity reduction of 95.27% and 86.35% compared to the DEP and EPA-IDD algorithms, respectively, while maintaining competitive BER performance. Additionally, the proposed DEPA algorithm provides a 0.84 dB performance gain over the MMSE-SDD algorithm without significant complexity overhead.

For $N_t = 16$, the DEPA algorithm can achieve 87.59% and 69.55% complexity reduction compared to the DEP and EPA-IDD algorithms, respectively, without any performance loss. In this antenna ratio, the mDEPA algorithm demonstrates performance similar to the DEPA algorithm while saving 37.3% of the complexity compared to DEPA algorithm.

V. CONCLUSION

We propose a low complexity DEPA algorithm for parallel detection and decoding in coded MIMO systems. Initially, we perform the DEPA algorithm to replace the exponential operations in the likelihood domain. Moreover, our proposed parallel IDD framework decouples the data dependency between inner and outer loops to achieve iteration convergence. Within this framework, the matrix inversion in EP-based IDD algorithms can be eliminated to reduce the computational complexity further. Finally, simulation results show that the proposed algorithms achieve a 69.55%∼95.27% reduction in computational complexity compared to SOA DEP detectors without noticeable performance degradation.

REFERENCES

[1] J. Céspedes, P. M. Olmos, M. Sánchez-Fernández, and F. Perez-Cruz, “Expectation propagation detection for high-order high-dimensional MIMO systems,” *IEEE Trans. Commun.*, vol. 62, no. 8, pp. 2840–2849, Aug. 2014.

[2] Y. Ge, L. Liu, S. Huang, G. D. González, Y. L. Guan, and Z. Ding, “Low-complexity memory AMP detector for high-mobility MIMO-OTFS SCMA systems,” in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2023, pp. 807–812.

[3] X. Li, H. Wang, Y. Ge, X. Shen, and J. Zhao, “Message feedback interference cancellation aided UAMP iterative detector for OTFS systems,” *IEEE Wireless Commun. Lett.*, vol. 13, no. 4, pp. 924–928, Apr. 2024.

[4] X. Tan, Y.-L. Ueng, Z. Zhang, X. You, and C. Zhang, “A low-complexity massive MIMO detection based on approximate expectation propagation,” *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 7260–7272, Aug. 2019.

[5] X. Tan et al., “Approximate expectation propagation massive MIMO detector with weighted Neumann-series,” *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 68, no. 2, pp. 662–666, Feb. 2021.

[6] Y. Chen, S. Song, Z. Wang, and J. Lin, “An efficient massive MIMO detector based on approximate expectation propagation,” *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 31, no. 5, pp. 696–700, May 2023.

[7] Y. Ge, Q. Deng, G. D. González, Y. L. Guan, and Z. Ding, “OTFS signaling for SCMA with coordinated multi-point vehicle communications,” *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 9044–9057, Jul. 2023.

[8] C. Studer, S. Fateh, and D. Seethaler, “ASIC implementation of soft-input soft-output MIMO detection using MMSE parallel interference cancellation,” *IEEE J. Solid-State Circuits*, vol. 46, no. 7, pp. 1754–1765, Jul. 2011.

[9] S. Ahmed and S. Kim, “Efficient SIC-MMSE MIMO detection with three iterative loops,” *AEU Int. J. Electron. Commun.*, vol. 72, pp. 65–71, Feb. 2017.

[10] S. Chan and S. Kim, “Min-sum soft symbol estimation for iterative MMSE detection,” *IEEE Access*, vol. 8, pp. 226240–226247, 2020.

[11] I. Santos and J. J. Murillo-Fuentes, “Self and turbo iterations for MIMO receivers and large-scale systems,” *IEEE Wireless Commun. Lett.*, vol. 8, no. 4, pp. 1095–1098, Aug. 2019.

[12] J. J. Murillo-Fuentes, I. Santos, J. C. Aradillas, and M. Sánchez-Fernández, “A low-complexity double EP-based detector for iterative detection and decoding in MIMO,” *IEEE Trans. Commun.*, vol. 69, no. 3, pp. 1538–1547, Mar. 2021.

[13] X. Tan, W. Li, Z. Zhang, X. You, and C. Zhang, “An efficient approximate EP-based iterative detection and decoding for massive MIMO,” *IEEE Wireless Commun. Lett.*, vol. 13, no. 2, pp. 255–259, Feb. 2024.

[14] Y. Ge, Q. Deng, P. C. Ching, and Z. Ding, “OTFS signaling for uplink NOMA of heterogeneous mobility users,” *IEEE Trans. Commun.*, vol. 69, no. 5, pp. 3147–3161, May 2021.

[15] A. Balatsoukas-Stimming, M. B. Parizi, and A. Burg, “LLR-based successive cancellation list decoding of polar codes,” *IEEE Trans. Signal Process.*, vol. 63, no. 19, pp. 5165–5179, Oct. 2015.

[16] P. Trifonov, “Efficient design and decoding of polar codes,” *IEEE Trans. Commun.*, vol. 60, no. 11, pp. 3221–3227, Nov. 2012.

[17] R. P. Brent and P. Zimmermann, *Modern Computer Arithmetic*. Cambridge, U.K.: Cambridge Univ. Press, 2010.